TRAN QUANG DUYEN

AI Engineer · LLM / RAG · Computer Vision

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SUMMARY

Fresh IT graduate from FPT University, aiming to become an AI Engineer. I have built my own LLM applications (RAG, AI agents) with two priorities: **reliable answers**, evaluated on locked test sets and controlled by code-level checks, and **production readiness**, with FastAPI services that track latency and cost. I have also completed a computer vision project with OpenCV and YOLO, from data preparation and training to evaluating models on traffic-camera video.

EDUCATION & LANGUAGES

FPT University · Information Technology 2022 – 2026

English — reads and understands technical documentation.

SKILLS

AI / LLM: RAG, AI Agent (LangChain/LangGraph), Embedding, Vector Database, Prompt engineering
Computer vision: YOLO object detection, object tracking, OpenCV, face recognition
ML & evaluation: Python, PyTorch, data labelling & augmentation, Precision / Recall / mAP, MRR / Recall@k
Backend & tools: FastAPI, REST API, Docker, Git, CI/CD (GitHub Actions), MongoDB, PostgreSQL

PROJECTS

SentinelAI — Traffic violation detection from camera video 10/2026

Personal project · github.com/duyentq-aie/SentinelAI

Python · PyTorch · YOLO11 · OpenCV · FastAPI · Kaggle

- Built an end-to-end system that detects traffic violations in traffic-camera video: riders without helmets, motorbikes carrying more than 2 people and red-light running (traffic-light colour read with OpenCV, checked per lane); results stream live to a FastAPI web app.
- Trained YOLO11 on Kaggle (GPU) to detect helmet / no-helmet / motorcycle, and tracked each vehicle across frames so a violation is reported once per vehicle, not once per frame.
- Found that the original dataset split put frames of the same video into both train and test, which inflated the scores; re-split by video and reported honest results (**test mAP50 0.93**, no-helmet 0.84) with per-class metrics and a confusion matrix.
- Tested on video from a new camera and checked every alert by eye: the retrained model **cut false alarms by half** (32 → 16 per minute); added a review page where a person confirms or rejects each alert before it counts.

ShopMate — AI sales assistant for a fashion store 10/2026

Personal project · github.com/duyentq-aie/shopmate

Python · LangGraph · Gemini · Qdrant · bge-m3 · Saleor (GraphQL) · FastAPI · Langfuse · GitHub Actions

- Built a shopping agent that answers product, return-policy and order-status questions on a Saleor store (250 products); read-only access, max 8 steps, support tickets only after the customer confirms.
- Raised search accuracy for Vietnamese queries typed without accents by **+0.43 MRR** (95% CI 0.27–0.58) by fusing accent-folded keyword search with bge-m3 semantic vectors in Qdrant.
- On a locked test set run once (33 questions): **31/33** correct answers, intent routing **33/33**, 2.6 s median latency, ~US$0.9 per 1,000 conversations; a code-level guardrail blocks replies with a wrong price.
- Evaluated like an experiment: 104-question golden set, test split locked and run only once; GitHub Actions CI runs 1,600+ automated tests and secret scanning.

DevPrep — AI mock-interview platform 2026

Capstone project (team) · github.com/Usopp204/DevPrep · FPT University

.NET 8 · YARP · SQL Server · FastAPI · Gemini · ChromaDB · InsightFace · SpeechBrain · Deepgram · Docker

- Owned 3 of 10+ .NET 8 microservices (Interview, Evaluation, ASP.NET Core MVC web client) and co-owned the FastAPI AI service, behind a YARP gateway with per-service SQL Server databases, all in Docker Compose.
- Built the interview flow: questions picked by role, level and focus areas from a **34,678**-question bank; voice room with Deepgram speech-to-text, Piper text-to-speech, Gemini hints, skip, pause/resume and idle auto-advance.
- Built RAG-based grading: Gemini scores each answer against reference answers retrieved from ChromaDB (bge-m3) on a **50/30/20** rubric (knowledge / problem-solving / communication) based on O*NET data.
- Built biometric anti-cheating with InsightFace (ArcFace) face and SpeechBrain voice embeddings: a verification gate before each session (3 attempts) and per-question face checks that flag no-face or multiple faces.

WORK EXPERIENCE

Backend Developer Intern 10/2024 – 02/2025

BIWOCO SOFTWARE COMPANY LIMITED · Software for the Australian market · 4 months

Python · FastAPI · MongoDB · AWS · Microservices · REST API

- Developed async REST APIs with FastAPI and Pydantic, with MongoDB queries (pagination, filtering, projection).
- Shipped features and bug fixes in a microservice backend on AWS; kept OpenAPI/Swagger docs up to date.
- Attended English-language Agile stand-ups and sprint reviews with the Australian client team.

AWARDS & CERTIFICATES

- Top 15 — FPTU Can Tho Olympic AI 2026, "AI for Smart Mekong Delta" · FPT University Can Tho, 2026
- Practical AI Training Program · FPT University Can Tho, 07–08/2026